

Домашнее задание

1. Решите уравнение.

1. $x^3 + 5x^2 - 4x - 20 = 0$
2. $x^3 - 3x^2 - 16x + 48 = 0$
3. $x^3 + 2x^2 - 9x - 18 = 0$
4. $x^3 - 7x^2 - x + 7 = 0$
5. $x^3 + 4x^2 = 25x + 100$
6. $x^3 - 6x^2 = 4x - 24$
7. $x^3 + 3x^2 - 2x - 6 = 0$
8. $x^3 - 5x^2 + x - 5 = 0$
9. $2x^3 - x^2 - 18x + 9 = 0$
10. $3x^3 + 2x^2 - 12x - 8 = 0$
11. $4x^3 - 12x^2 - x + 3 = 0$
12. $9x^3 + 18x^2 - x - 2 = 0$
13. $5x^3 + 4x^2 - 20x - 16 = 0$
14. $x^3 + 3x^2 - 9x - 27 = 0$
15. $x^3 - 12 = 4x - 3x^2$
16. $x^3 - 5x + 2x^2 - 10 = 0$
17. $x(x^2 + 6x + 9) = 4(x + 3)$
18. $(x - 2)(x^2 + 5x + 4) = 10(x - 2)$
19. $x^3 + 8 = 2x(x + 2)$
20. $x^3 - 27 = 13x(x - 3)$
21. $x^4 + 3x^3 - 8x - 24 = 0$
22. $x^4 - 2x^3 + x - 2 = 0$

2. Найдите значение выражения.

23. $5^{0,36} \cdot 25^{0,32}$
24. $\frac{12^{3,4}}{3^{1,4} \cdot 4^{2,4}}$
25. $\frac{10^{4,7}}{2^{2,7} \cdot 5^{3,7}}$
26. $\frac{2^{3,5} \cdot 3^{5,5}}{6^{4,5}}$
27. $\frac{(2^{-3})^2 \cdot 16^2}{8^{-1}}$
28. $\frac{3^{-5} \cdot 9^4}{3}$
29. $(1\frac{1}{4})^{-2} \cdot 0,8^{-3}$
30. $(4,5 \cdot 10^{-3}) \cdot (2 \cdot 10^5)$
31. $\frac{(7^{1/3} \cdot 7^{1/4})^{12}}{7^6}$
32. $1,5^{1/4} \cdot 2^{1/2} \cdot 6^{3/4}$
33. $\sqrt[3]{25} \cdot \sqrt[6]{25}$

$$34. \sqrt[3]{12} \cdot \sqrt[3]{18}$$

$$35. \frac{\sqrt[4]{48}}{\sqrt[4]{3}}$$

$$36. \frac{\sqrt{2,8} \cdot \sqrt{4,2}}{\sqrt{0,24}}$$

$$37. \frac{(\sqrt[3]{3} \cdot \sqrt{3})^6}{3^4}$$

$$38. \sqrt{65^2 - 56^2}$$

$$39. \sqrt{3} \frac{6}{25} - \sqrt{1} \frac{11}{25}$$

$$40. (\sqrt{13} - \sqrt{5})(\sqrt{13} + \sqrt{5})$$

$$41. (\sqrt{18} - \sqrt{8})^2$$

$$42. \frac{\sqrt{75} - \sqrt{12}}{\sqrt{3}}$$

$$43. (\sqrt{11} - 3)^2 + 6\sqrt{11}$$

$$44. \frac{4}{\sqrt{5} - 1} - \sqrt{5}$$

$$45. \frac{(\sqrt{7} + \sqrt{3})^2}{5 + \sqrt{21}}$$

$$46. \sqrt{(2 - \sqrt{7})^2} + \sqrt{(3 - \sqrt{7})^2}$$

$$47. \sqrt{7 - 4\sqrt{3}} + \sqrt{3}$$

$$48. \sqrt{11 + 6\sqrt{2}} - \sqrt{2}$$

$$49. \sqrt[3]{-0,125} + \sqrt[4]{0,0016}$$

$$50. \frac{5^{2\sqrt{3} + 1}}{25^{\sqrt{3}}}$$

$$51. \frac{a^{3,33}}{a^{2,11} \cdot a^{0,22}} \text{ при } a = 7$$

$$52. \frac{(b^{\sqrt{7}})^{\sqrt{7}}}{b^5} \text{ при } b = 3$$

$$53. \frac{x^{-4} \cdot (x^3)^3}{x^6} \text{ при } x = -2$$

$$54. \frac{(3a^2)^3 \cdot a^{-4}}{9a^2} \text{ при } a = 0,37$$

$$55. \frac{\sqrt[9]{a} \cdot \sqrt[18]{a}}{a \cdot \sqrt[6]{a}} \text{ при } a = 0,25$$

$$56. \frac{\sqrt{m}}{\sqrt[6]{m} \cdot \sqrt[3]{m}} \text{ при } m > 0$$

$$57. \frac{\sqrt[3]{x^2} \cdot \sqrt[6]{x}}{\sqrt{x}} \text{ при } x = 125$$

$$58. \frac{(\sqrt[5]{a} \cdot \sqrt[3]{b})^{15}}{a^2 b^4} \text{ при } a = 2, b = 3$$

$$59. \frac{\sqrt[4]{16a^8 b^4}}{a^2 b} \text{ при } b > 0$$